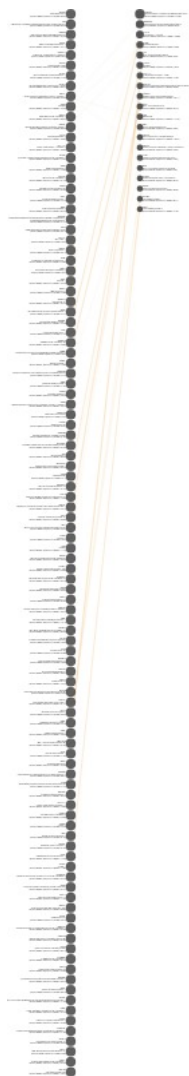


GeneMANIA report

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Application version : 3.6.0



Networks

- Co-expression
- Physical Interactions
- Genetic Interactions
- Predicted
- Co-localization
- Pathway
- Shared protein domains

Functions

N/A

Search parameters

Organism Homo sapiens (human)

Genes CFLAR , COPZ2 , RALA , NFIX , BIRC3 , IFNGR1 , DTNBP1 , CDK17 , MTHFD2 , TRNT1 , MGLL , SEMA3A , XPO1 , SNX10 , ZNF268 , HMOX1 , TNRC6B , MIB1 , CRISPLD2 , NIPAL2 , HPS5 , SLC11A2 , METAP2 , ARSB , NR3C1 , EHHADH , ARMC8 , FRMD4B , ATP6V1A , FOXP1 , EPHA4 , RALGPS2 , GBP3 , BAZ2B , AHDC1 , BEX2 , TCEAL4 , SORT1 , EMP1 , LRRC8A , MYC , THBS1 , FGD4 , TAF4B , ITGB3BP , DUSP10 , MBOAT2 , SOX13 , PHLDB2 , LPP , DDIT4L , USP53 , PIK3R1 , SHPRH , SNX30 , ZEB1 , COMMD7 , SLC7A11 , NEK7 , EPG5 , UTRN , APOOL , PI4K2A , RRAGA , SV2A , CAMK2N1 , DDR2 , PRICKLE2 , APBB2 , CASP3 , SYTL3 , GEM , SMCO4 , PLEKHA7 , MAP1A , TBC1D2B , FILIP1L , ADORA2B , PFKFB3 , FOXN2 , NUDT6 , PKIA , BCL2L1 , ENC1 , CERS6 , RAPH1 , GK5 , RPS6KA3 , TGIF1 , RPLP2 , CPNE7 , PER1 , GAS1 , KCTD2 , UPP1 , CSF1 , SNN , ZBTB37 , IRS2 , GSAP , GCNT1 , ARL4C , SFT2D2 , TXNIP

Network weighting Automatically selected weighting method

Networks A

Abbasi-Schild-Poulter-2019 , Abu-Odeh-Aqeilan-2014 , Achuthankutty-Mailand-2019 , Agrawal-Sedivy-2010 , Ahn-Lee-2008 , Albers-Koegl-2005 , Alexander-Wang-2018 , Alexandru-Deshaies-2008 , Alizadeh-Staudt-2000 , Alsulami-Cagney-2019 , An-Sun-2017 , Andresen-Flores-Morales-2014 , Arbogast-Gros-2019 , Arijs-Rutgeerts-2009 , Arroyo-Aloy-2014 , Arroyo-Aloy-2015 , Asadi-Dhanvantari-2018

B

Bailey-Hieter-2015 , Bandyopadhyay-Ideker-2010 , Banks-Washburn-2016 , Bantscheff-Drewes-2011 , Barr-Knapp-2009 , Barreiro-Alonso-Cerdán-2018 , Barrios-Rodiles-Wrana-2005 , Behrends-Harper-2010 , Behzadnia-Lührmann-2007 , Benleulmi-Chaachoua-Jockers-2016 A , Benleulmi-Chaachoua-Jockers-2016 B , Bennett-Harper-2010 , Benzinger-Hermeking-2005 , Berggård-James-2006 , Bett-Hay-2013 , Beyer-Boldt-2018 , Bhatnagar-Attie-2014 , Bild-Nevins-2006 B , BIOGRID-SMALL-SCALE-STUDIES , BIOGRID-SMALL-SCALE-STUDIES , Bishof-Seyfried-2018 , Blandin-Richard-2013 , Blomen-Brummelkamp-2015 , Blomen-Brummelkamp-2015 , Bogachek-Weigel-2014 , Boldrick-Relman-2002 , Boldt-Roepman-2016 , Botham-Schimmer-2019 , Bouwmeester-Superti-Furga-2004 , Brady-Omary-2018 , Brajenovic-Drewes-2004 , Brehme-Superti-Furga-2009 , Burington-Shaughnessy-2008 , Butland-Hayden-2014 , Byron-Humphries-2012

C

Cai-Conaway-2007 , Camargo-Brandon-2007 , Campos-Reinberg-2015 , Cao-Chinnaiyan-2014 , Carmon-Liu-2014 , Caron-van Attikum-2019 , CELL_MAP ,

C

Chen-Brown-2002 , Chen-Ge-2013 A , Chen-Ge-2013 B , Chen-Guan-2018 , Chen-Huang-2014 , Chen-Krogan-2018 , Chen-Yu-2018 , Chen-Zhang-2013 , Chen-Zhou-2019 , Cheng-DeCaprio-2017 , Chi-Reed-2018 , Chitale-Richly-2017 , Choi-Beutler-2019 , Choi-Busino-2018 , Choudhury-Michlewski-2017 , Christianson-Kopito-2011 , Cloutier-Coulombe-2013 , Cloutier-Coulombe-2017 , Colicelli-2010 , Colland-Gauthier-2004 , Conte-Perez-Oliva-2018 , Cooper-Green-2015 , Corominas-Iakoucheva-2014 , Couzens-Gingras-2013 , Cox-Rizzino-2013 , Coyaoud-Raught-2015 , Crow-Cristea-2017

D

Daakour-Twizere-2016 , Dabbaghizadeh-Tanguay-2018 , Dart-Wells-2015 , Das-Broemer-2019 , Davis-Glaunsinger-2015 , de Hoog-Mann-2004 , Devarajan-Ketha-Kumar-2012 , Diner-Cristea-2015 , Dittmer-Misteli-2014 , Dobbin-Giordano-2005 , Douanne-Bidère-2019 , Drissi-Boisvert-2015 , Du-Krogan-2017

E

Elliott-Gyrd-Hansen-2016 , Emdal-Olsen-2015 , Enzo-Dupont-2015 , Ertych-Bastians-2016 , Ewing-Figeys-2007

F

Fang-Lin-2011 , Faust-Frankel-2018 , Fenner-Prehn-2010 , Floyd-Pagliarini-2016 , Foerster-Ritter-2013 , Fogeron-Lange-2013 , Fonseca-Damgaard-2015 , Foster-Marshall-2013 , Fragoza-Yu-2019 , Freibaum-Taylor-2010

G

Gabriel-Baumgrass-2016 , Gallardo-Vara-Bernabeu-2019 , Galligan-Howley-2015 , Gao-Reinberg-2012 , Gao-Vaziri-2016 , Garzia-Sonenberg-2017 , Gautier-Hall-2009 , Giannone-Liu-2010 , Gilmore-Washburn-2016 , Giurato-Tarallo-2018 , Glatte-Gstaiger-2009 , Gloeckner-Ueffing-2007 , Goehler-Wanker-2004 , Gordon-Krogan-2020 , Goudreault-Gingras-2009 , Greco-Cristea-2011 , Grossmann-Stelzl-2015 , Guarani-Harper-2014 , Guard-Old-2019 , Guardia-Laguarta-Przedborski-2019 , Guderian-Grimmler-2011 , Gupta-Pelletier-2015

H

Han-Bassik-2017 A , Han-Bassik-2017 B , Hanson-Clayton-2014 , Hauri-Beisel-2016 , Hauri-Gstaiger-2013 , Havrylov-Redowicz-2009 , Havugimana-Emili-2012 , Hayes-Urbé-2012 , Hegele-Stelzl-2012 A , Hegele-Stelzl-2012 B , Heidelberger-Beli-2018 , Hein-Mann-2015 , Hermjakob-Apweiler-2004 , Herr-Helleday-2015 , Hoffmeister-Längst-2017 , Horlbeck-Gilbert-2018 A , Horlbeck-Gilbert-2018 B , Hosp-Selbach-2015 , Hou-Chen-2018 , Hou-Huang-2017 , Hu-Woods-2019 , Hu-Yin-2019 , Hubel-Pichlmair-2019 , Huber-Hoelz-2017 , HUMANCYC , Humphries-Humphries-2009 , Hussain-Aldaz-2018 , Hutchins-Peters-2010 , Huttlin-Gygi-2015 , Huttlin-Harper-2017 , Hüttenhain-Krogan-2019

I

I

I2D-BIND-Fly2Human , I2D-BIND-Mouse2Human , I2D-BIND-Rat2Human , I2D-BIND-Worm2Human , I2D-BIND-Yeast2Human , I2D-BioGRID-Fly2Human , I2D-BioGRID-Mouse2Human , I2D-BioGRID-Rat2Human , I2D-BioGRID-Worm2Human , I2D-BioGRID-Yeast2Human , I2D-Chen-Pawson-2009-PiwiScreen-Mouse2Human , I2D-Formstecher-Daviet-2005-Embryo-Fly2Human , I2D-Formstecher-Daviet-2005-Head-Fly2Human , I2D-Giot-Rothbert-2003-High-Fly2Human , I2D-Giot-Rothbert-2003-Low-Fly2Human , I2D-INNATEDB-Mouse2Human , I2D-IntAct-Fly2Human , I2D-IntAct-Mouse2Human , I2D-IntAct-Rat2Human , I2D-IntAct-Worm2Human , I2D-IntAct-Yeast2Human , I2D-Krogan-Greenblatt-2006-Core-Yeast2Human , I2D-Krogan-Greenblatt-2006-NonCore-Yeast2Human , I2D-Li-Vidal-2004-CE-DATA-Worm2Human , I2D-Li-Vidal-2004-CORE-1-Worm2Human , I2D-Li-Vidal-2004-CORE-2-Worm2Human , I2D-Li-Vidal-2004-interolog-Worm2Human , I2D-Li-Vidal-2004-literature-Worm2Human , I2D-Li-Vidal-2004-non-core-Worm2Human , I2D-Manual-Mouse2Human , I2D-Manual-Rat2Human , I2D-MGI-Mouse2Human , I2D-MINT-Fly2Human , I2D-MINT-Mouse2Human , I2D-MINT-Rat2Human , I2D-MINT-Worm2Human , I2D-MINT-Yeast2Human , I2D-MIPS-Yeast2Human , I2D-Ptacek-Snyder-2005-Yeast2Human , I2D-Stanyon-Finley-2004-CellCycle-Fly2Human , I2D-Tarassov-PCA-Yeast2Human , I2D-Tewari-Vidal-2004-TGFb-Worm2Human , I2D-vonMering-Bork-2002-High-Yeast2Human , I2D-vonMering-Bork-2002-Low-Yeast2Human , I2D-vonMering-Bork-2002-Medium-Yeast2Human , I2D-Wang-Orkin-2006-EScmplx-Mouse2Human , I2D-Wang-Orkin-2006-EScmplxIP-Mouse2Human , I2D-Wang-Orkin-2006-EScmplxlow-Mouse2Human , I2D-Yu-Vidal-2008-GoldStd-Yeast2Human , IMID , Ingham-Pawson-2005 , Innocenti-Brown-2011 , INTERPRO , Iradi-Borchelt-2018 , IREF-bhf-ucl , IREF-bind , IREF-bind-translation , IREF-biogrid , IREF-corum , IREF-dip , IREF-hpidb , IREF-hprd , IREF-huri , IREF-innatedb , IREF-intact , IREF-intcomplex , IREF-matrixdb , IREF-mbinfo , IREF-mint , IREF-mppi , IREF-quickgo , IREF-reactome , IREF-SMALL-SCALE-STUDIES , IREF-SMALL-SCALE-STUDIES , IREF-spike , IREF-uniprotpp , IREF-virushost , Ivanochko-Arrowsmith-2019

J

Jain-Parker-2016 , Jang-Trono-2018 , Jeronimo-Coulombe-2007 , Jiang-de Kok-2017 , Jin-Pawson-2004 , Jirawatnotai-Sicinski-2011 , Johnson-Kerner-Wichterle-2015 , Johnson-Shoemaker-2003 , Jones-MacBeath-2006 , Joshi-Cristea-2013 , Jozwik-Carroll-2016 , Jäger-Krogan-2011

K

Kahle-Zoghbi-2011 , Kaltenbach-Hughes-2007 , Kang-Shin-2015 , Karras-Soengas-2019 , Kato-Sternberg-2014 , Katsogiannou-Rocchi-2014 , Kawahara-Paes Leme-2017 , Keller-Lee-2014 , Kennedy-Kolch-2020 A , Kennedy-Kolch-2020 B , Khanna-Parnaik-2018 , Kim-Major-2015 , Kneissl-Grummt-2003 , Koch-Hermeking-2007 , Kotlyar-Jurisica-2015 , Kristensen-Foster-2012 , Kumar-Maddika-2017 , Kumar-Vertegaal-2017 , Kupka-Walczak-2016 , Kärblane-Sarmiento-2015 , Kırılı-Görlich-

K

2015

L

Lambert-Gingras-2015 , Lampert-Peter-2018 , Lau-Ronai-2012 , Lee-Choi-2016 , Lee-Choi-2017 , Lee-Jeong-2017 , Lee-Jou-2019 , Lee-Mayr-2019 , Lee-Songyang-2011 , Lehner-Sanderson-2004 A , Lehner-Sanderson-2004 B , Leung-Jones-2014 , Leung-Miller-2017 , Li-Chen-2015 , Li-Dorf-2011 A , Li-Dorf-2011 B , Li-Dorf-2014 , Li-Fu-2017 , Li-Haura-2013 , Li-Hung-2019 , Li-Lu-2018 , Li-Wang-2016 , Li-Zhou-2017 , Liebelt-Vertegaal-2020 , Lim-Zoghbi-2006 , Lin-Smith-2010 , Lipp-Guthrie-2015 , Liu-Chen-2019 , Liu-Sun-2019 , Liu-Takahashi-2017 , Liu-Tan-2018 , Liu-Varjosalo-2018 , Liu-Wang-2012 , Liu-Xu-2018 , Liu-Yang-2019 , Llères-Lamond-2010 , Loch-Strickler-2012 , Low-Heck-2014 , Lu-Bohr-2017 , Lu-Zhang-2013 , Luck-Calderwood-2020 , Lum-Cristea-2018 , Luo-Elledge-2009

M

Mak-Moffat-2010 , Malinová-Verheggen-2017 , Mallon-McKay-2013 , Malovannaya-Qin-2010 , Maly-Babu-2017 , Markson-Sanderson-2009 , Martin-Elledge-2017 , Maréchal-Zou-2014 , Matsumoto-Nakayama-2005 , Matsuoka-Elledge-2007 , McCracken-Blencowe-2005 , McFarland-Nussbaum-2008 , McNamara-D'Orso-2016 , Meek-Piwnica-Worms-2004 , Menon-Litovchick-2019 , Milev-Mouland-2012 , Miyamoto-Sato-Yanagawa-2010 , Mohammed-Carroll-2013 , Moon-Kim-2014 , Moutaoufik-Babu-2019 , Mugabo-Lim-2018 , Muller-Demeret-2012 , Murakawa-Landthaler-2015

N

Nakamura-Groth-2019 , Nakayama-Ohara-2002 , Napolitano-Meroni-2011 , Narayan-Bennett-2012 , Nassa-Weisz-2019 , Nathan-Goldberg-2013 , NCI_NATURE , Neganova-Lako-2011 , Newman-Keating-2003 , Noguchi-Kawahara-2018 , Nowak-Sommer-2019

O

Oliviero-Cagney-2015 , Oliviero-Cagney-2016 , Olma-Pintard-2009 , Oláh-Ovádi-2011 , Ouyang-Gill-2009

P

Panigrahi-Pati-2012 , Pankow-Yates-2015 , Pao-Virdee-2018 , Papp-Lamia-2015 , Pech-Settleman-2019 , Perez-Hernandez-Yáñez-Mó-2013 , Perez-Perri-Espinosa-2016 , Perou-Botstein-1999 , Perou-Botstein-2000 , Persaud-Rotin-2009 A , Persaud-Rotin-2009 B , Petschnigg-Stagljar-2014 , PFAM , Phillips-Corn-2013 , Pichlmair-Superti-Furga-2011 , Pichlmair-Superti-Furga-2012 , Pilling-Cooper-2017 , Pladevall-Morera-Lopez-Contreras-2019 , Ptushkina-Ray-2017

R

Raisner-Gascoigne-2018 , Ramachandran-LaBaer-2004 , Raman-Harper-2015 , Ramaswamy-Golub-2001 , Ravasi-Hayashizaki-2010 , REACTOME , Reinke-

R

Keating-2010 , Reinke-Keating-2013 , Rengasamy-Walsh-2017 , Reyniers-Taymans-2014 , Richter-Chrzanowska-Lightowlers-2010 , Rieger-Chu-2004 , Rivera-Paes Leme-2018 , Rodriguez-von Kriegsheim-2016 , Roewenstrunk-de la Luna-2019 , Rolland-Vidal-2014 , Rosenbluh-Hahn-2016 , Rosenwald-Staudt-2001 , Ross-Perou-2001 , Roth-Zlotnik-2006 , Rowbotham-Mermoud-2011 , Roy-Pardo-2014 , Roy-Parent-2013 , Rual-Vidal-2005

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Saez-Vilchez-2018 , Sahni-Vidal-2015 , Saito-Kobarg-2017 , Sala-Ampe-2017 , Salvetti-Greco-2016 , Sang-Jackson-2011 , Sato-Conaway-2004 , Savidis-Brass-2016 , Schadt-Shoemaker-2004 , Schiza-Diamandis-2018 , Scholz-Taylor-2016 , Scifo-Lalowski-2015 , Scott-Guy-2017 , Scott-Schulman-2016 , Shami Shah-Baskin-2019 , Shen-Chen-2019 , Shen-Mali-2017 , Sherman-Teitell-2010 , Simabuco-Zanchin-2019 , Singh-Moore-2012 , So-Colwill-2015 , Sokolina-Stagljär-2017 , Soler-López-Aloy-2011 , Sowa-Harper-2009 , Srivas-Ideker-2016 , St-Denis-Gingras-2015 , St-Denis-Gingras-2016 , Stehling-Lill-2012 , Stehling-Lill-2013 , Stelzl-Wanker-2005 , Stuart-Kim-2003 , Sundell-Ivarsson-2018 , Suter-Wanker-2013 , Swayampakula-Dedhar-2017

T

Taipale-Lindquist-2012 , Taipale-Lindquist-2014 , Takahashi-Conaway-2011 , Tang-Wang-2019 , Tarallo-Weisz-2011 , Teixeira-Gomes-2010 , Teixeira-Laman-2016 A , Teixeira-Laman-2016 B , Thalappilly-Duseti-2008 , Thompson-Luchansky-2014 , Tiemann-Kani-2019 , Tomkins-Manzoni-2018 , Tong-Moran-2014 , Toyoshima-Grandori-2012 , Trepte-Wanker-2018 A , Trepte-Wanker-2018 B , Tsai-Cristea-2012

U

Ugidos-Vandenbroeck-2019

V

Van Acker-Dewilde-2019 , Van Alstyne-Pellizzoni-2018 , Van Quickelberghe-Gevaert-2018 , van Wijk-Timmers-2009 , Vandamme-Angrand-2011 , Varier-Vermeulen-2016 , Varjosalo-Gstaiger-2013 A , Varjosalo-Gstaiger-2013 B , Varjosalo-Supert-Furga-2013 , Vastrik-Stein-2007 , Venkatesan-Vidal-2009 , Viita-Vartiainen-2019 , Vinayagam-Wanker-2011 , Virok-Fülöp-2011 , Vizeacoumar-Moffat-2013 , von Hundelshausen-Weber-2017

W

Wallach-Kramer-2013 , Wan-Emili-2015 , Wang-Balch-2006 , Wang-Cheung-2015 , Wang-He-2008 , Wang-Huang-2017 , Wang-Liu-2019 , Wang-Maris-2006 , Wang-Xiong-2019 , Wang-Xu-2015 , Wang-Yang-2011 , Watanabe-Fujita-2018 , Weimann-Stelzl-2013 A , Weimann-Stelzl-2013 B , Weinmann-Meister-2009 , Weishäupl-Schmidt-2019 , Weith-Meyer-2018 , Whisenant-Salomon-2015 , Wilkinson-Coba-2019 , Willingham-Muchowski-2003 , Winczura-Jensen-2018 , Wong-O'Bryan-2012 , Woods-Monteiro-2012 A , Woods-Monteiro-2012 B , Woodsmith-Sanderson-2012 ,

W

Wu-Garvey-2007 , Wu-Li-2007 , Wu-Ma-2012 , Wu-Stein-2010 , Wu-Stein-2010

X

Xiao-Brown-2018 , Xiao-Lefkowitz-2007 , Xie-Cong-2013 , Xie-Green-2012 , Xie-Zhang-2017 , Xu-Ye-2012 , Xu-Zetter-2016

Y

Yachie-Roth-2016 , Yadav-Varjosalo-2017 , Yamauchi-Maeda-2018 , Yang-Brasier-2015 , Yang-Chen-2010 , Yang-Maurer-2018 , Yang-Vidal-2016 , Yang-Wang-2018 , Yao-Stagljar-2017 A , Yao-Stagljar-2017 B , Yatim-Benkirane-2012 , Yeung-Dougan-2019 , Yu-Chow-2013 , Yu-Engel-2018 , Yu-Vidal-2011 , Yue-Liu-2018

Z

Zanon-Pichler-2013 , Zeller-Wei-2006 , Zhang-Shang-2006 , Zhang-Vermeulen-2017 , Zhang-Wang-2018 , Zhang-Wheeler-2014 , Zhang-Xu-2018 , Zhang-Zou-2011 , Zhao-Krug-2005 , Zhao-Yang-2011 , Zhong-Vidal-2016 , Zhou-Conrads-2004 , Zhou-Hanemann-2016 , Zhu-Liu-2018

Genes

Gene	Description	Rank
COMMD7	COMM domain containing 7 [Source:HGNC Symbol;Acc:HGNC:16223]	N/A
SNX30	sorting nexin family member 30 [Source:HGNC Symbol;Acc:HGNC:23685]	N/A
GK5	glycerol kinase 5 [Source:HGNC Symbol;Acc:HGNC:28635]	N/A
PHLDB2	pleckstrin homology like domain family B member 2 [Source:HGNC Symbol;Acc:HGNC:29573]	N/A
DDIT4L	DNA damage inducible transcript 4 like [Source:HGNC Symbol;Acc:HGNC:30555]	N/A
PER1	period circadian regulator 1 [Source:HGNC Symbol;Acc:HGNC:8845]	N/A
EPG5	ectopic P-granules autophagy protein 5 homolog [Source:HGNC Symbol;Acc:HGNC:29331]	N/A
ZBTB37	zinc finger and BTB domain containing 37 [Source:HGNC Symbol;Acc:HGNC:28365]	N/A
RAPH1	Ras association (RalGDS/AF-6) and pleckstrin homology domains 1 [Source:HGNC Symbol;Acc:HGNC:14436]	N/A
PLEKHA7	pleckstrin homology domain containing A7 [Source:HGNC Symbol;Acc:HGNC:27049]	N/A
GBP3	guanylate binding protein 3 [Source:HGNC Symbol;Acc:HGNC:4184]	N/A
PRICKLE2	prickle planar cell polarity protein 2 [Source:HGNC Symbol;Acc:HGNC:20340]	N/A
TRNT1	tRNA nucleotidyl transferase 1 [Source:HGNC Symbol;Acc:HGNC:17341]	N/A
SYTL3	synaptotagmin like 3 [Source:HGNC Symbol;Acc:HGNC:15587]	N/A
SHPRH	SNF2 histone linker PHD RING helicase [Source:HGNC Symbol;Acc:HGNC:19336]	N/A
FGD4	FYVE, RhoGEF and PH domain containing 4 [Source:HGNC Symbol;Acc:HGNC:19125]	N/A
DTNBP1	dystrobrevin binding protein 1 [Source:HGNC Symbol;Acc:HGNC:17328]	N/A
LRRC8A	leucine rich repeat containing 8 VRAC subunit A [Source:HGNC Symbol;Acc:HGNC:19027]	N/A
FRMD4B	FERM domain containing 4B [Source:HGNC Symbol;Acc:HGNC:24886]	N/A
APOOL	apolipoprotein O like [Source:HGNC Symbol;Acc:HGNC:24009]	N/A
NIPAL2	NIPA like domain containing 2 [Source:HGNC Symbol;Acc:HGNC:25854]	N/A
MIB1	mindbomb E3 ubiquitin protein ligase 1 [Source:HGNC Symbol;Acc:	N/A

Gene	Description	Rank
	HGNC:21086]	
RALGPS2	Ral GEF with PH domain and SH3 binding motif 2 [Source:HGNC Symbol;Acc:HGNC:30279]	N/A
COPZ2	COPI coat complex subunit zeta 2 [Source:HGNC Symbol;Acc:HGNC:19356]	N/A
NUDT6	nudix hydrolase 6 [Source:HGNC Symbol;Acc:HGNC:8053]	N/A
FOXP1	forkhead box P1 [Source:HGNC Symbol;Acc:HGNC:3823]	N/A
SFT2D2	SFT2 domain containing 2 [Source:HGNC Symbol;Acc:HGNC:25140]	N/A
RPLP2	ribosomal protein lateral stalk subunit P2 [Source:HGNC Symbol;Acc:HGNC:10377]	N/A
SMCO4	single-pass membrane protein with coiled-coil domains 4 [Source:HGNC Symbol;Acc:HGNC:24810]	N/A
CRISPLD2	cysteine rich secretory protein LCCL domain containing 2 [Source:HGNC Symbol;Acc:HGNC:25248]	N/A
USP53	ubiquitin specific peptidase 53 [Source:HGNC Symbol;Acc:HGNC:29255]	N/A
BEX2	brain expressed X-linked 2 [Source:HGNC Symbol;Acc:HGNC:30933]	N/A
CERS6	ceramide synthase 6 [Source:HGNC Symbol;Acc:HGNC:23826]	N/A
SNX10	sorting nexin 10 [Source:HGNC Symbol;Acc:HGNC:14974]	N/A
ARMC8	armadillo repeat containing 8 [Source:HGNC Symbol;Acc:HGNC:24999]	N/A
CAMK2N1	calcium/calmodulin dependent protein kinase II inhibitor 1 [Source:HGNC Symbol;Acc:HGNC:24190]	N/A
SOX13	SRY-box transcription factor 13 [Source:HGNC Symbol;Acc:HGNC:11192]	N/A
APBB2	amyloid beta precursor protein binding family B member 2 [Source:HGNC Symbol;Acc:HGNC:582]	N/A
SNN	stannin [Source:HGNC Symbol;Acc:HGNC:11149]	N/A
PI4K2A	phosphatidylinositol 4-kinase type 2 alpha [Source:HGNC Symbol;Acc:HGNC:30031]	N/A
CSF1	colony stimulating factor 1 [Source:HGNC Symbol;Acc:HGNC:2432]	N/A
CPNE7	copine 7 [Source:HGNC Symbol;Acc:HGNC:2320]	N/A
MAP1A	microtubule associated protein 1A [Source:HGNC Symbol;Acc:HGNC:6835]	N/A
TBC1D2B	TBC1 domain family member 2B [Source:HGNC Symbol;Acc:HGNC:29183]	N/A
CDK17	cyclin dependent kinase 17 [Source:HGNC Symbol;Acc:HGNC:8750]	N/A
ARL4C	ADP ribosylation factor like GTPase 4C [Source:HGNC Symbol;Acc:	N/A

Gene	Description	Rank
	HGNC:698]	
AHDC1	AT-hook DNA binding motif containing 1 [Source:HGNC Symbol;Acc:HGNC:25230]	N/A
SORT1	sortilin 1 [Source:HGNC Symbol;Acc:HGNC:11186]	N/A
SEMA3A	semaphorin 3A [Source:HGNC Symbol;Acc:HGNC:10723]	N/A
SLC11A2	solute carrier family 11 member 2 [Source:HGNC Symbol;Acc:HGNC:10908]	N/A
EMP1	epithelial membrane protein 1 [Source:HGNC Symbol;Acc:HGNC:3333]	N/A
ZNF268	zinc finger protein 268 [Source:HGNC Symbol;Acc:HGNC:13061]	N/A
FOXP2	forkhead box N2 [Source:HGNC Symbol;Acc:HGNC:5281]	N/A
NFIX	nuclear factor I X [Source:HGNC Symbol;Acc:HGNC:7788]	N/A
KCTD2	potassium channel tetramerization domain containing 2 [Source:HGNC Symbol;Acc:HGNC:21294]	N/A
EPHA4	EPH receptor A4 [Source:HGNC Symbol;Acc:HGNC:3388]	N/A
CFLAR	CASP8 and FADD like apoptosis regulator [Source:HGNC Symbol;Acc:HGNC:1876]	N/A
METAP2	methionyl aminopeptidase 2 [Source:HGNC Symbol;Acc:HGNC:16672]	N/A
MBOAT2	membrane bound O-acyltransferase domain containing 2 [Source:HGNC Symbol;Acc:HGNC:25193]	N/A
SLC7A11	solute carrier family 7 member 11 [Source:HGNC Symbol;Acc:HGNC:11059]	N/A
TNRC6B	trinucleotide repeat containing adaptor 6B [Source:HGNC Symbol;Acc:HGNC:29190]	N/A
GAS1	growth arrest specific 1 [Source:HGNC Symbol;Acc:HGNC:4165]	N/A
GCNT1	glucosaminyl (N-acetyl) transferase 1 [Source:HGNC Symbol;Acc:HGNC:4203]	N/A
TAF4B	TATA-box binding protein associated factor 4b [Source:HGNC Symbol;Acc:HGNC:11538]	N/A
IRS2	insulin receptor substrate 2 [Source:HGNC Symbol;Acc:HGNC:6126]	N/A
ARSB	arylsulfatase B [Source:HGNC Symbol;Acc:HGNC:714]	N/A
SV2A	synaptic vesicle glycoprotein 2A [Source:HGNC Symbol;Acc:HGNC:20566]	N/A
LPP	LIM domain containing preferred translocation partner in lipoma [Source:HGNC Symbol;Acc:HGNC:6679]	N/A
NEK7	NIMA related kinase 7 [Source:HGNC Symbol;Acc:HGNC:13386]	N/A
PFKFB3	6-phosphofructo-2-kinase/fructose-2,6-biphosphatase 3 [Source:HGNC	N/A

Gene	Description	Rank
	Symbol;Acc:HGNC:8874]	
UTRN	utrophin [Source:HGNC Symbol;Acc:HGNC:12635]	N/A
GSAP	gamma-secretase activating protein [Source:HGNC Symbol;Acc:HGNC:28042]	N/A
DDR2	discoidin domain receptor tyrosine kinase 2 [Source:HGNC Symbol;Acc:HGNC:2731]	N/A
BAZ2B	bromodomain adjacent to zinc finger domain 2B [Source:HGNC Symbol;Acc:HGNC:963]	N/A
ATP6V1A	ATPase H ⁺ transporting V1 subunit A [Source:HGNC Symbol;Acc:HGNC:851]	N/A
RPS6KA3	ribosomal protein S6 kinase A3 [Source:HGNC Symbol;Acc:HGNC:10432]	N/A
HMOX1	heme oxygenase 1 [Source:HGNC Symbol;Acc:HGNC:5013]	N/A
ENC1	ectodermal-neural cortex 1 [Source:HGNC Symbol;Acc:HGNC:3345]	N/A
TCEAL4	transcription elongation factor A like 4 [Source:HGNC Symbol;Acc:HGNC:26121]	N/A
FILIP1L	filamin A interacting protein 1 like [Source:HGNC Symbol;Acc:HGNC:24589]	N/A
HPS5	HPS5 biogenesis of lysosomal organelles complex 2 subunit 2 [Source:HGNC Symbol;Acc:HGNC:17022]	N/A
ADORA2B	adenosine A2b receptor [Source:HGNC Symbol;Acc:HGNC:264]	N/A
DUSP10	dual specificity phosphatase 10 [Source:HGNC Symbol;Acc:HGNC:3065]	N/A
THBS1	thrombospondin 1 [Source:HGNC Symbol;Acc:HGNC:11785]	N/A
PKIA	cAMP-dependent protein kinase inhibitor alpha [Source:HGNC Symbol;Acc:HGNC:9017]	N/A
MTHFD2	methylenetetrahydrofolate dehydrogenase (NADP ⁺ dependent) 2, methenyltetrahydrofolate cyclohydrolase [Source:HGNC Symbol;Acc:HGNC:7434]	N/A
TGIF1	TGFB induced factor homeobox 1 [Source:HGNC Symbol;Acc:HGNC:11776]	N/A
RALA	RAS like proto-oncogene A [Source:HGNC Symbol;Acc:HGNC:9839]	N/A
ZEB1	zinc finger E-box binding homeobox 1 [Source:HGNC Symbol;Acc:HGNC:11642]	N/A
UPP1	uridine phosphorylase 1 [Source:HGNC Symbol;Acc:HGNC:12576]	N/A
GEM	GTP binding protein overexpressed in skeletal muscle [Source:HGNC Symbol;Acc:HGNC:4234]	N/A
RRAGA	Ras related GTP binding A [Source:HGNC Symbol;Acc:HGNC:16963]	N/A

Gene	Description	Rank
BCL2L1	BCL2 like 1 [Source:HGNC Symbol;Acc:HGNC:992]	N/A
TXNIP	thioredoxin interacting protein [Source:HGNC Symbol;Acc:HGNC:16952]	N/A
MGLL	monoglyceride lipase [Source:HGNC Symbol;Acc:HGNC:17038]	N/A
EHHADH	enoyl-CoA hydratase and 3-hydroxyacyl CoA dehydrogenase [Source:HGNC Symbol;Acc:HGNC:3247]	N/A
CASP3	caspase 3 [Source:HGNC Symbol;Acc:HGNC:1504]	N/A
IFNGR1	interferon gamma receptor 1 [Source:HGNC Symbol;Acc:HGNC:5439]	N/A
ITGB3BP	integrin subunit beta 3 binding protein [Source:HGNC Symbol;Acc:HGNC:6157]	N/A
PIK3R1	phosphoinositide-3-kinase regulatory subunit 1 [Source:HGNC Symbol;Acc:HGNC:8979]	N/A
BIRC3	baculoviral IAP repeat containing 3 [Source:HGNC Symbol;Acc:HGNC:591]	N/A
NR3C1	nuclear receptor subfamily 3 group C member 1 [Source:HGNC Symbol;Acc:HGNC:7978]	N/A
XPO1	exportin 1 [Source:HGNC Symbol;Acc:HGNC:12825]	N/A
MYC	MYC proto-oncogene, bHLH transcription factor [Source:HGNC Symbol;Acc:HGNC:7553]	N/A
ATRAID	all-trans retinoic acid induced differentiation factor [Source:HGNC Symbol;Acc:HGNC:24090]	1
HNRNPU	heterogeneous nuclear ribonucleoprotein U [Source:HGNC Symbol;Acc:HGNC:5048]	2
CYCS	cytochrome c, somatic [Source:HGNC Symbol;Acc:HGNC:19986]	3
DTNB	dystrobrevin beta [Source:HGNC Symbol;Acc:HGNC:3058]	4
AXL	AXL receptor tyrosine kinase [Source:HGNC Symbol;Acc:HGNC:905]	5
CCND1	cyclin D1 [Source:HGNC Symbol;Acc:HGNC:1582]	6
COL11A1	collagen type XI alpha 1 chain [Source:HGNC Symbol;Acc:HGNC:2186]	7
MAP4K4	mitogen-activated protein kinase kinase kinase kinase 4 [Source:HGNC Symbol;Acc:HGNC:6866]	8
VANGL2	VANGL planar cell polarity protein 2 [Source:HGNC Symbol;Acc:HGNC:15511]	9
MAX	MYC associated factor X [Source:HGNC Symbol;Acc:HGNC:6913]	10
TXN	thioredoxin [Source:HGNC Symbol;Acc:HGNC:12435]	11
NMI	N-myc and STAT interactor [Source:HGNC Symbol;Acc:HGNC:7854]	12
XRCC5	X-ray repair cross complementing 5 [Source:HGNC Symbol;Acc:HGNC:12833]	13

Gene	Description	Rank
NR2F6	nuclear receptor subfamily 2 group F member 6 [Source:HGNC Symbol;Acc:HGNC:7977]	14
CCN1	cellular communication network factor 1 [Source:HGNC Symbol;Acc:HGNC:2654]	15
BAD	BCL2 associated agonist of cell death [Source:HGNC Symbol;Acc:HGNC:936]	16
RAD9A	RAD9 checkpoint clamp component A [Source:HGNC Symbol;Acc:HGNC:9827]	17
NGFR	nerve growth factor receptor [Source:HGNC Symbol;Acc:HGNC:7809]	18
RALBP1	ralA binding protein 1 [Source:HGNC Symbol;Acc:HGNC:9841]	19
YBX3	Y-box binding protein 3 [Source:HGNC Symbol;Acc:HGNC:2428]	20

Networks

Co-expression	55.14%
Chen-Brown-2002	5.96%
Gene expression patterns in human liver cancers. Chen et al (2002). <i>Mol Biol Cell</i> Co-expression with 291,300 interactions from supplementary material	
Wang-Cheung-2015	5.62%
Genetic variation in insulin-induced kinase signaling. Wang et al (2015). <i>Mol Syst Biol</i> Co-expression with 422,896 interactions from GEO	
Bild-Nevins-2006 B	4.05%
Oncogenic pathway signatures in human cancers as a guide to targeted therapies. Bild et al (2006). <i>Nature</i> Co-expression with 285,368 interactions from GEO	
Jiang-de Kok-2017	3.62%
Omics-based identification of the combined effects of idiosyncratic drugs and inflammatory cytokines on the development of drug-induced liver injury. Jiang et al (2017). <i>Toxicol Appl Pharmacol</i> Co-expression with 444,959 interactions from GEO	
Innocenti-Brown-2011	3.60%
Identification, replication, and functional fine-mapping of expression quantitative trait loci in primary human liver tissue. Innocenti et al (2011). <i>PLoS Genet</i> Co-expression with 620,205 interactions from GEO	
Wang-Maris-2006	3.43%
Integrative genomics identifies distinct molecular classes of neuroblastoma and shows that multiple genes are targeted by regional alterations in DNA copy number. Wang et al (2006). <i>Cancer Res</i> Co-expression with 270,388 interactions from GEO	
Arijs-Rutgeerts-2009	3.16%
Mucosal gene expression of antimicrobial peptides in inflammatory bowel disease before and after first infliximab treatment. Arijs et al (2009). <i>PLoS One</i> Co-expression with 676,695 interactions from GEO	
Dobbin-Giordano-2005	3.07%
Interlaboratory comparability study of cancer gene expression analysis using oligonucleotide microarrays. Dobbin et al (2005). <i>Clin Cancer Res</i> Co-expression with 452,322 interactions from GEO	
Mallon-McKay-2013	2.91%
StemCellDB: the human pluripotent stem cell database at the National Institutes of Health. Mallon et al (2013). <i>Stem Cell Res</i> Co-expression with 602,113 interactions from GEO	
Ramaswamy-Golub-2001	2.79%
Multiclass cancer diagnosis using tumor gene expression signatures. Ramaswamy et al (2001). <i>Proc Natl Acad Sci U S A</i> Co-expression with 284,829 interactions from supplementary material	
Burington-Shaughnessy-2008	2.79%
Tumor cell gene expression changes following short-term in vivo exposure to single agent chemotherapeutics are related to survival in multiple myeloma. Burington et al (2008). <i>Clin Cancer Res</i>	

Co-expression	55.14%
Burington-Shaughnessy-2008	
Co-expression with 295,320 interactions from GEO	
Perou-Botstein-1999	2.30%
Distinctive gene expression patterns in human mammary epithelial cells and breast cancers. Perou et al (1999). <i>Proc Natl Acad Sci U S A</i>	
Co-expression with 68,200 interactions from supplementary material	
Rieger-Chu-2004	2.23%
Toxicity from radiation therapy associated with abnormal transcriptional responses to DNA damage. Rieger et al (2004). <i>Proc Natl Acad Sci U S A</i>	
Co-expression with 266,879 interactions from GEO	
Wu-Garvey-2007	1.93%
The effect of insulin on expression of genes and biochemical pathways in human skeletal muscle. Wu et al (2007). <i>Endocrine</i>	
Co-expression with 275,155 interactions from GEO	
Alizadeh-Staudt-2000	1.86%
Distinct types of diffuse large B-cell lymphoma identified by gene expression profiling. Alizadeh et al (2000). <i>Nature</i>	
Co-expression with 92,360 interactions from supplementary material	
Ross-Perou-2001	1.46%
A comparison of gene expression signatures from breast tumors and breast tissue derived cell lines. Ross et al (2001). <i>Dis Markers</i>	
Co-expression with 146,858 interactions from supplementary material	
Rosenwald-Staudt-2001	1.26%
Relation of gene expression phenotype to immunoglobulin mutation genotype in B cell chronic lymphocytic leukemia. Rosenwald et al (2001). <i>J Exp Med</i>	
Co-expression with 118,097 interactions from supplementary material	
Perou-Botstein-2000	1.25%
Molecular portraits of human breast tumours. Perou et al (2000). <i>Nature</i>	
Co-expression with 189,373 interactions from supplementary material	
Boldrick-Relman-2002	1.06%
Stereotyped and specific gene expression programs in human innate immune responses to bacteria. Boldrick et al (2002). <i>Proc Natl Acad Sci U S A</i>	
Co-expression with 116,197 interactions from supplementary material	
Roth-Zlotnik-2006	0.82%
Gene expression analyses reveal molecular relationships among 20 regions of the human CNS. Roth et al (2006). <i>Neurogenetics</i>	
Co-expression with 683,844 interactions from GEO	
Physical Interactions	32.18%
Stelzl-Wanker-2005	5.57%
A human protein-protein interaction network: a resource for annotating the proteome. Stelzl et al (2005). <i>Cell</i>	
Physical Interactions with 3,370 interactions from iRefIndex	
IREF-mppi	5.43%
Physical Interactions with 304 interactions from iRefIndex	
	4.38%

Physical Interactions	32.18%
Kırlı-Görlich-2015	
A deep proteomics perspective on CRM1-mediated nuclear export and nucleocytoplasmic partitioning. Kırlı et al (2015). <i>Elife</i>	
Physical Interactions with 1,034 interactions from BioGRID	
Zeller-Wei-2006	3.36%
Global mapping of c-Myc binding sites and target gene networks in human B cells. Zeller et al (2006). <i>Proc Natl Acad Sci U S A</i>	
Physical Interactions with 634 interactions from iRefIndex	
Agrawal-Sedivy-2010	2.60%
Proteomic profiling of Myc-associated proteins. Agrawal et al (2010). <i>Cell Cycle</i>	
Physical Interactions with 105 interactions from BioGRID	
Kumar-Maddika-2017	1.79%
A Human Tyrosine Phosphatase Interactome Mapped by Proteomic Profiling. Kumar et al (2017). <i>J Proteome Res</i>	
Physical Interactions with 1,863 interactions from BioGRID	
Sowa-Harper-2009	1.64%
Defining the human deubiquitinating enzyme interaction landscape. Sowa et al (2009). <i>Cell</i>	
Physical Interactions with 1,509 interactions from BioGRID	
IREF-uniprotpp	1.62%
Physical Interactions with 2,470 interactions from iRefIndex	
Wu-Li-2007	1.40%
Systematic identification of SH3 domain-mediated human protein-protein interactions by peptide array target screening. Wu et al (2007). <i>Proteomics</i>	
Physical Interactions with 1,105 interactions from iRefIndex	
Heidelberger-Beli-2018	0.98%
Proteomic profiling of VCP substrates links VCP to K6-linked ubiquitylation and c-Myc function. Heidelberger et al (2018). <i>EMBO Rep</i>	
Physical Interactions with 1,347 interactions from BioGRID	
Ptushkina-Ray-2017	0.92%
A non-transcriptional role for the glucocorticoid receptor in mediating the cell stress response. Ptushkina et al (2017). <i>Sci Rep</i>	
Physical Interactions with 142 interactions from BioGRID	
BIOGRID-SMALL-SCALE-STUDIES	0.74%
Physical Interactions with 79,201 interactions from BioGRID	
IREF-mint	0.63%
Physical Interactions with 14,408 interactions from iRefIndex	
Hein-Mann-2015	0.57%
A human interactome in three quantitative dimensions organized by stoichiometries and abundances. Hein et al (2015). <i>Cell</i>	
Physical Interactions with 27,015 interactions from BioGRID	
Lee-Mayr-2019	0.42%
Gain of Additional BIRC3 Protein Functions through 3'-UTR-Mediated Protein Complex Formation. Lee et al (2019). <i>Mol Cell</i>	
Physical Interactions with 1,759 interactions from BioGRID	
IREF-bind	0.14%

Physical Interactions	32.18%
IREF-bind	
Physical Interactions with 3,524 interactions from iRefIndex	
IREF-quickgo	0.00%
Physical Interactions with 9,985 interactions from iRefIndex	
Genetic Interactions	4.09%
Lin-Smith-2010	3.00%
A genome-wide map of human genetic interactions inferred from radiation hybrid genotypes. Lin et al (2010). <i>Genome Res</i>	
Genetic Interactions with 4,805,334 interactions from supplementary material	
Toyoshima-Grandori-2012	1.09%
Functional genomics identifies therapeutic targets for MYC-driven cancer. Toyoshima et al (2012). <i>Proc Natl Acad Sci U S A</i>	
Genetic Interactions with 101 interactions from BioGRID	
Predicted	3.86%
I2D-IntAct-Rat2Human	2.18%
The IntAct molecular interaction database in 2010. Aranda et al (2010). <i>Nucleic Acids Res</i>	
Predicted with 1,690 interactions from I2D	
I2D-BioGRID-Yeast2Human	0.77%
BioGRID: a general repository for interaction datasets. Stark et al (2006). <i>Nucleic Acids Res</i>	
Predicted with 17,314 interactions from I2D	
I2D-BioGRID-Rat2Human	0.71%
BioGRID: a general repository for interaction datasets. Stark et al (2006). <i>Nucleic Acids Res</i>	
Predicted with 2,148 interactions from I2D	
Wu-Stein-2010	0.19%
A human functional protein interaction network and its application to cancer data analysis. Wu et al (2010). <i>Genome Biol</i>	
Predicted with 89,967 interactions from supplementary material	
I2D-BioGRID-Mouse2Human	0.01%
BioGRID: a general repository for interaction datasets. Stark et al (2006). <i>Nucleic Acids Res</i>	
Predicted with 10,524 interactions from I2D	
Co-localization	2.67%
Johnson-Shoemaker-2003	2.36%
Genome-wide survey of human alternative pre-mRNA splicing with exon junction microarrays. Johnson et al (2003). <i>Science</i>	
Co-localization with 426,464 interactions from GEO	
Schadt-Shoemaker-2004	0.30%
A comprehensive transcript index of the human genome generated using microarrays and computational approaches. Schadt et al (2004). <i>Genome Biol</i>	
Co-localization with 59,920 interactions from GEO	
Pathway	1.99%
IMID	1.63%
Pathway with 1,023 interactions from Pathway Commons	

Pathway	1.99%
NCI_NATURE	0.36%
Pathway with 10,118 interactions from Pathway Commons	
Shared protein domains	0.06%
INTERPRO	0.06%
Shared protein domains with 621,159 interactions from InterPro	